

individual stocks. Individual stocks are exposed to the market factor, which carries the risk premium (it is the nutrient), but also have *idiosyncratic risk*, which is not rewarded by a risk premium (this is the part that carries no nutritional value). Investors can diversify away the idiosyncratic part and increase their returns by holding the market factor portfolio, rather than any other combination of individual stocks. The market portfolio represents *systematic risk*, and it is pervasive: all risky assets have risk premiums determined only by their exposure to the market portfolio. Market risk also affects all investors, except those who are infinitely risk averse and hold only risk-free assets.

The key to this result is diversification. The CAPM is based on investors having mean-variance utility and, as chapter 3 shows, the most important concept in mean-variance investing is diversification. Diversification ensures that, absent perfect correlation, when one asset performs badly, some other assets will perform well, and so gains can partly offset losses. Investors never want to hold assets in isolation; they improve their risk–return trade-off by diversifying and holding portfolios of many assets. This balance across many assets that are not perfectly correlated improves Sharpe ratios. Investors will diversify more and more until they hold the most diversified portfolio possible—the market portfolio. The market factor is the best, most-well diversified portfolio investors can hold under the CAPM.

The CAPM states that the market portfolio is held by every investor—a strong implication that is outright rejected in data. Nevertheless, it is useful to understand how we can leap from a diversified portfolio to the market being the only relevant factor.

Recall the mean-variance frontier with the capital allocation line (CAL) from chapter 3 (see Figure 3.10), which is reproduced in Figure 6.3. This is the solution to the mean-variance investing problem. Investors hold different amounts of the risk-free asset and the mean-variance efficient (MVE) portfolio depending on their risk aversion. Now here come the strong assumptions of the CAPM. Assume that the set of means, volatilities, and correlations are the same for all investors.

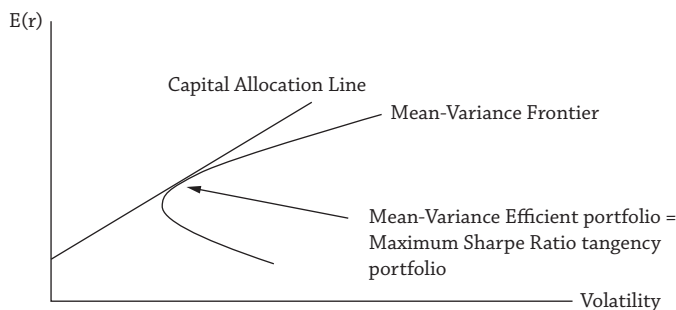


Figure 6.3